

1

Trading Mechanics and Market Structure

Learning objectives:

- How securities markets are organized
- Who sets the rules
- How the organization of securities markets has changed recently

1.1 Introduction

Securities markets are mechanisms for bringing buyers and sellers together and enabling them to trade. Trading may be prompted by various factors: the need to mitigate risks (hedging), the desire to exploit superior information (speculation), or the urge to rebalance one's portfolio (liquidity shocks). In standard treatments of asset pricing, such as the capital asset pricing model (CAPM), the trading mechanism is not laid out explicitly, on the assumption that it does not matter for securities prices. Yet in reality there is a wide variety of trading mechanisms, and market participants pay close attention to their design. Changes in trading rules are often hotly debated, and market organizers carefully fine-tune these rules to improve the competitiveness of their trading platform. This is because the trading rules affect the efficiency of markets as mechanisms to realize trading gains and discover asset values. They also affect the apportioning of gains among market participants, determining, for instance, the fraction of the gain that is captured by specialized intermediaries.

A trading mechanism defines the “rules of the game” that market participants must follow: it determines the actions they can take (e.g., the kinds of orders they can place), their information about other market participants' actions (e.g., whether they observe quotes or orders), and the protocol for matching buy and sell orders (e.g., whether orders are executed at a common price or not). As the possible rules can be put together in a virtually boundless number of combinations, real-world market structures feature great diversity and are constantly

evolving, so attempting a complete classification is hopeless. It is more fruitful to focus on two prototype trading mechanisms, namely the limit order market (or auction market) and the dealer market.¹ In fact, all securities markets rely, in various ways, on combinations of these two basic mechanisms. Section 1.2 describes how each operates, illustrates market structures that combine elements of both, and shows that each prototypical mechanism itself can vary in important ways, such as the degree of transparency and the frequency of trades.

In limit order markets, all participants (final investors and intermediaries) can interact directly and multilaterally on one trading platform; their bids and offers are consolidated in a limit order book (LOB) according to price priority, so that higher bids and cheaper offers are more likely to be executed. As all participants can trade with each other, trading platforms using this mechanism are sometimes referred to as “all-to-all” trading systems. As explained below, such trading platforms are used in equity and derivatives markets (e.g., options and futures).

By contrast, in dealer markets final investors can only trade at the bid and ask quotes posted by specialized intermediaries, called “dealers” or “market makers,” and these quotes are not consolidated to enforce price priority. As prices are often set via bilateral negotiations between dealers and investors (or their brokers), these markets are often referred to as “Over the Counter” (OTC), an old expression referring to decentralized sales, generally occurring over the counter of a shop. Fixed income securities (e.g., Treasuries, corporate bonds, or interest rate swaps), currencies, and interbank loans are often traded in this way. As explained in Section 1.2.2, OTC markets have experienced important changes in recent years (due to automation and regulation) with the emergence of electronic trading systems enabling investors to trade directly with each other (“all-to-all” trading systems) or to request quotes from multiple dealers at the same time (“Request for Quotes” systems or RFQs). Moreover, dealers now use electronic limit order books to trade among each other, in so-called interdealer markets.

Market design matters. Indeed, much of this book distills the results of the large body of research on how market design affects trading costs and price discovery. To provide an idea of the impact that the design of security markets can have on their performance, Section 1.3 briefly previews some empirical studies that compare limit order and dealer markets or investigate markets with different degrees of transparency, i.e., where markets provide different amounts of information about the trading process.

¹ These two basic market structures are also known, respectively, as “order-driven” and “quote-driven” markets.

An obvious question is why in practice markets feature different trading mechanisms. As Section 1.4 explains, trading rules are determined by the interplay between regulators, intermediaries, issuers, investors, and the managers of trading platforms. The balance between these stakeholders—and hence the actual trading rules—largely depends on the governance and ownership of the platform. For instance, the platform may be managed for profit or not, and the ownership shares of the various stakeholders (intermediaries, issuers, investors) may vary considerably between platforms and over time. The design of a trading platform must also take into account the possible threat from competing platforms—a concern that in recent decades has become more pressing due to a combination of capital market liberalization, changes in security regulation, and technological advances. In particular, digital and communication technology has radically transformed the trading process, sparking an increasingly lively debate on the impact of new trading technologies on market liquidity, price volatility, and economic efficiency.

1.2 Limit Order Markets and Dealer Markets

1.2.1 Limit Order Markets

A defining feature of the limit order market is that buy and sell orders from all market participants are matched directly in a single marketplace, which can be either the floor of an exchange or a “trading platform” run by a computer (see Figure 1.1).² Orders go into a Limit Order Book (LOB), which determines the

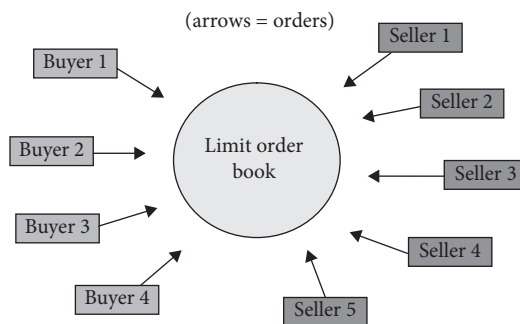


Figure 1.1 Limit order market.

² However, even when trading occurs on such trading platforms, it may not be completely centralized, as multiple limit order books for the same asset may coexist. As explained in Section 1.4.2 and Chapter 7, such market fragmentation occurs for instance in equity markets in North America and Europe.

priority with which they will be matched with offsetting orders, according to the rules of the market and the characteristics of the orders themselves. In call (or batch) markets, incoming orders are stored in the LOB and then matched at discrete intervals, such as once per day. In continuous markets, they are matched immediately with orders already present on the LOB, if possible; otherwise they are stored in the LOB to await future execution.

Continuous Limit Order Markets

When submitting orders to a continuous limit order market, investors can design them differently depending on their trading needs. The most basic choice, which determines both speed and price of execution, is between limit and market orders. A buy limit order specifies the maximum price at which the trader is prepared to buy a stated amount of the security; similarly, a sell limit order specifies the minimum price the seller will accept for a given amount. A market order only specifies an amount to buy or sell, not the price: it will therefore be executed at whatever price it can fetch on the market.

Limit orders may not find a counterpart with which they can be matched at the specified price. Market orders, by contrast, are filled immediately if there is any outstanding limit order on the other side of the market. Thus, one difference between limit and market orders is that limit orders do not guarantee immediate execution—indeed, they may never be executed at all—whereas market orders are executed immediately upon submission.

The LOB shown in Figure 1.2 illustrates the mechanics of trading in a continuous limit order market. The LOB is a snapshot at a given point in time of all the limit orders awaiting execution. In the LOB, buy limit orders (bids) are arranged in decreasing order of price and sell limit orders (asks) in increasing order. The LOB also shows size and time, that is, the number of shares specified and the time the order was entered in the book. Depending on the market's trading rules, only a subset of the orders present in the LOB may be visible to market participants. As we shall see later, this is one dimension of market transparency.

Consider an investor who wants to buy nine hundred shares. He has two options. One is to place a buy market order for this amount. In this case, the order is executed immediately against the best limit orders to sell (on the ask side): it will first fill the two limit orders placed at the offer price of \$74.48 for eight hundred shares, and then be executed for the remaining hundred shares against the limit order at \$75.74, so that its average execution price is \$74.62. The order of priority in which limit orders on the book are executed depends on their price: aggressively priced orders are filled before less competitive ones. In other words, execution obeys a price priority rule. If two limit orders have the same price, they are filled according to secondary priority rules such as time

Market sell order of 200 (or
limit sell with price < 74.42)

Market buy order of 900 (or
limit buy with price > 75.74)

Bid			Ask		
Price	Size	Time	Price	Size	Time
74.42	300 100	11:49:39	74.48	300	11:49:35
74.41	100	11:46:55	74.48	500	11:49:40
74.36	400	11:48:30	75.74	100	08:25:17
74.36	400	11:48:32	76.00	150	08:02:02
74.00	13	10:56:00	76.77	20	07:01:01
73.75	5100	11:28:02	77.00	100	09:15:00
72.98	5100	10:56:99	77.06	200	10:14:11
72.15	120	08:01:39	77.35	1000	08:01:39
72.11	20	07:01:01	77.82	20	07:01:01
72.03	20	07:01:01	78.00	300	08:02:00
72.00	Because of the buy market order, the bid-ask spread widens from $74.48 - 74.42 = 0.06$ to $76.00 - 74.42 = 1.58$. The market order has “consumed” liquidity.				9:30:04
71.59					8:01:32
71.11					9:30:04
71.00					7:01:01
70.35					8:01:35
70.11	20	07:01:01	80.00	350	09:15:00

Figure 1.2 Example of limit order book (LOB).

of submission or pro-rata allocation (fractional execution proportional to limit order size).

The second option for the investor is to place a buy limit order for nine hundred shares. If he specifies a limit price lower than \$74.48, the order is entered in the LOB on the bid side and stored for future execution. The level of the chosen bid price determines the likelihood and speed of execution, as more aggressively priced buy orders are executed first according to price priority.

If the investor instead specifies a limit price equal to or higher than \$74.48—that is, if he matches or crosses the best price on the ask side of the limit book—then the order is *marketable*: it can be executed at once, at least partially, against stored sell limit orders, in this example those at \$74.48. If the order specifies a limit price of \$74.50, the remaining hundred shares will appear on the bid side of the LOB as a buy limit order at \$74.50. Significantly, the transaction price (\$74.48) is determined by existing prices on the LOB, not by the price of the incoming marketable limit order.

The treatment of sellers is analogous. For instance, an investor who wants to sell two hundred shares immediately can either place a market sell order for two hundred shares or a marketable limit order with a price of \$74.42 or less. If he is more patient, he can improve his execution by placing a sell limit order for two hundred shares at a price above \$74.42, on the ask side of the market. But in this case he runs the risk of non-execution.

Hence, the choice between a market and a limit order involves a trade-off between immediate execution at current market prices and a more favorable transaction price at the cost of delayed and uncertain execution. This trade-off is studied in detail in Chapter 6.

Figure 1.2 can also be used to illustrate the concept of illiquidity, which is discussed in Chapter 2. As a thought experiment, consider a “round-trip transaction,” that is, a buy market order followed by an equal-size sell market order. If the market were perfectly liquid, the cost of this round-trip transaction would be zero. Instead, the figure shows that it has a positive cost that increases with its size. If the order size is smaller than three hundred, one buys at \$74.48 and resells at \$74.42, so that the round-trip cost is \$0.06 (i.e., six cents). This cost—the difference between the best bid and the best ask price on the market—is called the “quoted bid-ask spread” and is often used as a measure of illiquidity.

For larger orders, one can compute a similar measure of illiquidity by comparing the average price paid by a buyer placing a large market order and the average price received by a seller for an equally large order. The buy price rises with order size, because the buyer has to “walk up” the schedule of sell limit orders to fill his own buy order. Symmetrically, the sell price is decreasing with the size of the order, as the seller has to “walk down” the schedule of buy limit orders. Thus larger orders are associated with a greater difference between the average execution price for buy and sell market orders—the “weighted average bid-ask spread.”³ A market in which investors can trade large quantities without substantially moving the price—that is, where the weighted average bid-ask spread does not increase much with trade size—is said to be “deep.” Therefore, *market depth* is inversely related to the weighted average spread for large trade size. The concept of depth will be made more precise in Chapter 4.

The LOB evolves in real time as market and limit orders are submitted and earlier limit orders are cancelled. For instance, in our example, the submission of a buy market order for nine hundred shares depletes the LOB on the ask side and so widens the bid-ask spread from 0.06 to 1.58. By contrast, if the buyer submits a limit order at \$74.45, the bid-ask spread narrows to 0.03. Since market orders widen the spread, they are viewed as consuming liquidity, and traders

³ For instance, in the LOB depicted in Figure 1.2, only eight hundred shares can be bought at \$74.48: a buyer looking for one thousand shares would also have to buy one hundred shares at \$75.74 and one hundred more at \$76.00. As a result, the average price paid per share would be $(74.48 \times 0.8) + (75.74 \times 0.1) + (76.00 \times 0.1) = 74.76$. Conversely, one thousand shares can be sold at an average price of $(74.42 \times 0.3) + (74.41 \times 0.1) + (74.36 \times 0.6) = 74.38$. Therefore, the weighted average bid-ask spread for one thousand shares is $74.76 - 74.38 = 0.38$, rather than 0.06. As a percentage of the mid-quote, the bid-ask spread rises from 0.08% to 0.5%.

submitting these orders are called “liquidity demanders” (or liquidity takers). In contrast, those submitting limit orders are called “liquidity suppliers” (or liquidity makers), since limit orders replenish the LOB.

Box 1.1 Other Types of Order

Limit and market orders are by far the most common kinds of order. But trading platforms often also allow for more complex orders. A *stop order* is an instruction to buy (or sell) only once the price has risen to (fallen to, respectively) a certain level. A stop sell order can be used to limit one's losses on holding a stock if its price nosedives. Moreover, traders can set conditions on cancellation with their orders: *good-until-cancel* orders are valid until they are cancelled, while *good-until* orders are valid until a specified date and *immediate-or-cancel* (or *fill-or-kill*) orders are valid only at the moment they reach the market. Finally, *hidden orders* are limit orders that are stored in the LOB but not displayed to market participants. These orders will be executed in the same way as regular limit orders, but they usually lose time priority against limit orders that are displayed at the same price. A variant of the hidden limit order is the so-called iceberg order, for which a fraction of the actual size is shown to other market participants along with the price. As the order is executed, the hidden size becomes gradually evident to market participants.

Call Limit Order Markets

So far we have considered limit order markets with continuous matching. Another arrangement matches orders at discrete points in time. In this case the limit order market is known as a call (or batch) auction. Before the call auction is held, market participants can submit market and limit orders over a pre-specified time window. During this window, orders gradually accumulate in the LOB without being filled and can in general be cancelled at any time. This pre-auction period stops at some time (which can be deterministic or random), when the price at which eligible orders are executed is set. Unlike in the continuous limit order market, in the call auction all executable orders are cleared at the same price. For this reason, the call auction is sometimes called a single or uniform price auction.

More precisely, at the time of the call auction, all the buy orders in hand are sorted in decreasing order of limit price, with buy market orders treated as at the highest possible price. This determines the cumulative quantity that traders

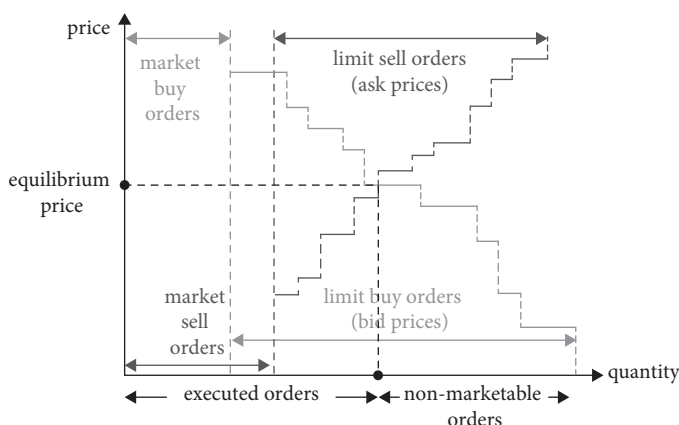


Figure 1.3 Call auction.

are prepared to buy at each possible price. Symmetrically, the sell orders are sorted by increasing limit price, with sell market orders treated as at the lowest possible price. This determines the cumulative quantity that would be sold at each possible price. The resulting demand and supply functions are the two stepwise schedules shown in Figure 1.3.

The price set in the auction—the market-clearing or equilibrium price—is determined by the point where these two stepwise schedules intersect. At this price, orders from all buyers with a bid higher than the clearing price and all sellers with a price below that clearing price are fully executed. Limit orders with a price just equal to the clearing price (the marginal traders) may be partially executed. For instance, in Figure 1.3, the total demand at the clearing price exceeds the total supply, so the marginal buyer will get only partial execution. Buy limit orders below the clearing price or sell orders above it are not filled. It is easy to see that the clearing price maximizes the (voluntary) trading volume, as it leaves no trading opportunity unexploited.

In the past, many exchanges in continental Europe were call markets. In fact, the economist Léon Walras was inspired by the mechanism of the Paris Bourse call auction when he formalized the process by which supply and demand are balanced in competitive markets. Thus, the notion of Walrasian market routinely presented in introductory economics courses is very close to the way trading is organized in call markets. Today, in equity markets, the call auction is often used to determine the opening price (before the start of continuous limit order trading) and the closing price. For instance, the NYSE, the Nasdaq, the Toronto Stock Exchange, Euronext, the London Stock Exchange (LSE), and the Italian Stock Exchange all use call auctions to set opening and closing prices.

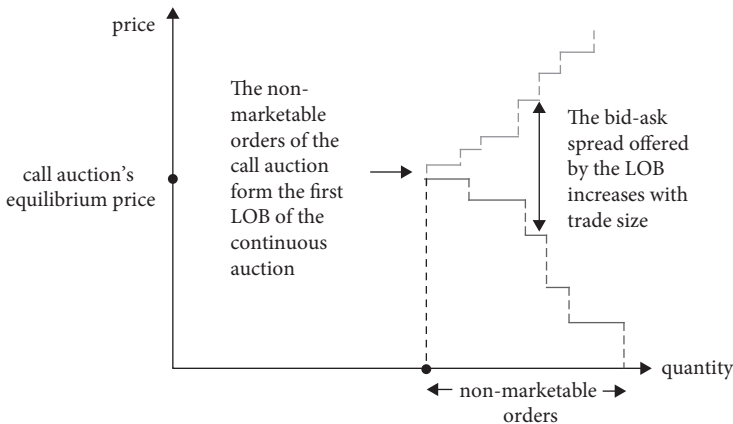


Figure 1.4 Forming the initial limit order book of the trading day.

At the opening, the limit orders that are unfilled in the opening call auction form the initial LOB for the continuous session. Figure 1.4 shows this initial LOB in the case of the call auction displayed in Figure 1.3 and illustrates that the bid-ask spread on the LOB increases with trade size, as noted.

Opening and closing auctions are important for different reasons. At the opening, call auctions play an important role in price discovery: the call auction helps to aggregate information received by different market participants while the market is closed overnight. Meanwhile, closing auctions are becoming increasingly important, as the closing price serves as a benchmark to value institutional investors' portfolios; moreover, as the closing auction attracts a lot of trading interest, it enables asset managers to rebalance their portfolios without the risk of large adverse impact on prices. For instance, for U.S. stocks, closing auctions accounted for about 8% of the daily trading volume in 2018 (Bogousslavsky and Muravyev 2021).

Call auctions are also sometimes used as the only trading mechanism for stocks that are traded infrequently. In this way, market organizers make sure that there is sufficient interest on both sides of the market. They increase the likelihood of finding a counterpart for each side while reducing the risk that the clearing price will be distorted by a temporary imbalance between supply and demand. For instance, on the LSE, SETSqx is a trading platform that runs four electronic auctions a day (alongside a dealer market) for securities that are less liquid than those traded on SETS. Some trading platforms (e.g., Cboe Europe) also run periodic call auctions during the trading, in parallel to the continuous limit order book.

As the trading rules are different in continuous limit order markets and call auctions, traders will not behave in the same way in each case. A formal analysis

of price formation and traders' behavior in call and continuous limit order markets is provided in Chapter 4 and Chapter 6, respectively.

Dark Pools

Dark pools are trading platforms that are accessible only to institutional investors (and sometimes financial intermediaries). These are electronic trading platforms operated by stock exchanges (e.g., Turquoise by the LSE, Smartpool by Euronext, or Xetra by the Deutsche Börse), brokers (e.g., BlockCross by ICAP or Blockmatch by Instinet) or banks (e.g., SigmaX by Goldman Sachs or SG CIB AlphaY by Société Générale). These platforms operate in parallel with continuous limit order markets and therefore offer investors an alternative way to execute their orders.

Generally, dark pools do not contribute to price discovery, to the extent that they simply operate as “crossing networks,” where traders submit buy and sell market orders and these are periodically crossed at reference prices drawn from other markets, such as the mid-point between the best bid and ask price on the main market for the stock. If there is excess demand (supply) at the reference price, some buyers (sellers) are rationed according to some rule (e.g., time priority or pro rata). Other types of dark pools (e.g., MatchNow, operated by Cboe for Canadian stocks) are not pure crossing networks, allowing traders to submit limit orders, insofar as these offer price improvements relative to the best quotes in the main market.

These trading mechanisms are “dark” in the sense that orders submitted to them are not displayed to the rest of market participants. Moreover, dark pools only disseminate aggregate information on trading volume after executing trades (without revealing information on unfilled orders, for instance). Institutional investors are attracted to dark pools precisely because their opacity reduces the risk of information leakage about their trading intentions (see Chapter 8). The attractiveness of dark pools has increased further in response to the rise of algorithmic trading (see Chapter 9).

Thus, it is no surprise that dark pools have attracted an increasing share of trading volume in recent years: for example, their share of European equity trading rose from less than 1% to about 8% between 2009 and 2016 (Petrescu and Wedow 2017); in U.S. equities markets, trading volume in dark pools rose to about 10% of total trading volume by 2019 (SEC 2020). The growth of dark pool trading—and more generally, off-exchange trading—has attracted regulatory attention: for instance, in Europe the recent MiFID 2 directive capped trading volume in dark pools at 8% of the total in each stock. More broadly, the growth of dark pool trading raises questions about market fragmentation (see Chapter 7) and the optimal level of transparency (see Section 1.2.5 and Chapter 8).

1.2.2 Dealer Markets

In dealer markets, the final investors (or their brokers) do not trade directly with each other, but must contact a dealer, find out his price, and trade at this price, or else try another dealer. So in a dealer market there is a sharp distinction between liquidity suppliers (the dealers) and liquidity demanders (final investors), whereas in a limit order market each participant chooses whether to provide or to demand liquidity.

Figures 1.5 and 1.6 illustrate the trading process in a dealer market. As an example, suppose that Seller 4 wants to sell sixty shares and that he first contacts

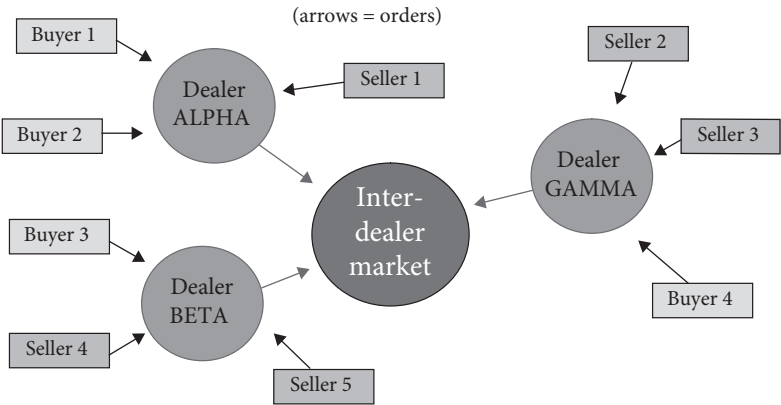


Figure 1.5 Dealer market.

2	ALPHA ZETA	326–329	GAMMA EPSILON IOTA	3
Market Maker Identity Code	Bid Price	Offer Price	Quote Size	Time of Latest Quote Update
ALPHA	326	330	75 × 75	08:53
BETA	324	330	75 × 75	09:14
GAMMA	325	329	75 × 75	09:16
DELTA	323	332	75 × 75	08:53
EPSILON	325	329	25 × 25	09:36
ZETA	326	330	75 × 75	11:30
ETA	325	330	75 × 75	09:45
THETA	325	330	75 × 75	09:23
IOTA	324	329	75 × 75	10:27
KAPPA	323	330	75 × 75	09:45
LAMBDA	325	330	75 × 75	08:53

Figure 1.6 Dealers bid and ask quotes.

dealer Beta. As shown in Figure 1.6, Beta is willing to buy at \$324 and sell at \$330. Seller 4 can then decide either to sell at \$324 or to seek another dealer. In the first case, Beta fills Seller 4's order by buying the security and adding it to his inventory. This exposes Beta to the risk of a sudden fall in the price of the security, and hence a loss on the value of his inventory. To avoid this "inventory risk," Beta can either rebalance his position by trading with a customer who wants to buy the security (for instance, Buyer 3 in the figure) or he can contact other dealers to sell them part or all of his position. As we will see in Chapter 3, the management of inventory risk is a major determinant of bid and ask prices.

Thus, we can distinguish two different segments in dealer markets (as shown in Figure 1.5): the Dealer-to-Customer (D2C) segment, in which dealers serve final investors, and the Dealer-to-Dealer (D2D) segment ("interdealer market"), in which dealers trade with each other to share inventory risk. To do so, dealers use specialized "interdealer brokers" (IDBs) such as ICAP, BGC Partners or BrokerTec (part of the CME group). These brokers have gradually introduced electronic limit order books enabling dealers to trade anonymously and quickly among themselves. Examples of such interdealer trading platforms include eSpeed or BrokerTec (introduced in the late 1990s) in U.S. Treasury markets, MTS (introduced in 1999) in E.U. sovereign bonds, or Thomson Reuters (now Refinitiv) and EBS (Electronic Broker Service) in currencies markets (introduced in 1992 and 1993, respectively).

The volume of trade on the D2D market is typically larger than on the D2C market, as each trade with a given client trickles down to other dealers (via "intermediation chains") until it is passed on to final investors on the opposite side of the market.⁴ Eisfeldt et al. (2023) document that in the market for single-name credit default swaps trading among fourteen core dealers accounts for 83% of the total market volume. However, for corporate bonds and currencies, the share of interdealer trading has declined over time, due to the electronification of trading in the D2C market and to regulation (see Section 1.4). For instance, in foreign exchange markets the share of interdealer trading has declined from about two-thirds of total trading volume in the early 1990s to less than one-half (see Chaboud et al. 2023).

In a dealer market, unlike a limit order market, there is no enforcement of price priority: in the example of Figure 1.5, Seller 4 trades with Beta even though he could obtain a better price from Alpha. This can happen because quotes are not necessarily displayed to final investors, who must therefore find the best price

⁴ For example, in the U.S. municipal bond markets, 23% of transactions initiated by customers generate intermediation chains that involve at least two dealers (see table III in Li and Schürhoff 2018).

by contacting dealers. This search is costly: it takes time and effort and, for this reason, it increases dealers' market power and illiquidity (see Section 10.4).

In contrast, when information on quotes is publicly available, market participants can easily identify the dealers who post the best bid and ask prices. Some dealer markets feature "Multi Dealer Platforms" (MDPs) that consolidate information on the quotes posted by dealers and disseminate it via screens that provide real-time information similar to that in Figure 1.6.⁵ Such consolidated information is important, as shown by the example presented in the figure: no single dealer quotes a bid-ask spread of less than 4, but the spread resulting from the consolidation of the quotes (sometimes called the "inside spread" or "market touch" in the United Kingdom) is $329 - 326 = 3$, as is shown at the top of the panel (which also tells us which, and how many, market makers quote the best price on each side of the market). Thus, the market as a whole offers more liquidity than any individual dealer.

Many dealer markets, however, offer far less detail on dealers' quotes. For instance, no real-time information is available in OTC markets such as the U.S. or the European corporate bond market. In currency markets, Reuters and Bloomberg screens do give information on quotes, but this is only indicative: the quotes do not commit dealers to actually trade at those prices.

Dealers' quotes are typically valid only for a limited number of shares. So a large order may be executed by splitting it among several dealers. Suppose that a seller wishes to sell three hundred shares given the dealers' quotes in Figure 1.6. He can execute this order by selling seventy-five shares each to dealers Alpha and Zeta, who post the best bid price, and then another seventy-five each to Gamma and Lambda at the next best bid price. Effectively, the investor is walking down the demand curve resulting from the aggregation of dealers' bid quotes. Similarly, a buyer with a large order will walk up the aggregate supply curve resulting from the dealers' ask quotes. These aggregate demand and supply curves are shown in Figure 1.7. Therefore, as in a limit order market, in a dealer market one can also define a weighted-average bid-ask spread that is also increasing in trade size.

Unlike limit order markets, dealer markets often enable traders to bargain over price and quantity. For instance, instead of searching for a better price, Seller 4 in our example could ask Beta for a better price than 324. If Beta agrees, then Seller 4 gets what is called a *price improvement*. These are common in some dealer markets and result in trades at prices within the quoted bid-ask spread. Moreover, by design, dealer markets allow dealers to establish long-term relationships with their clients and other dealers. They can therefore reward

⁵ MDPs (such as 360T and SWAPS) have been developed first in foreign exchange markets but are now also used for other asset classes that trade over the counter.

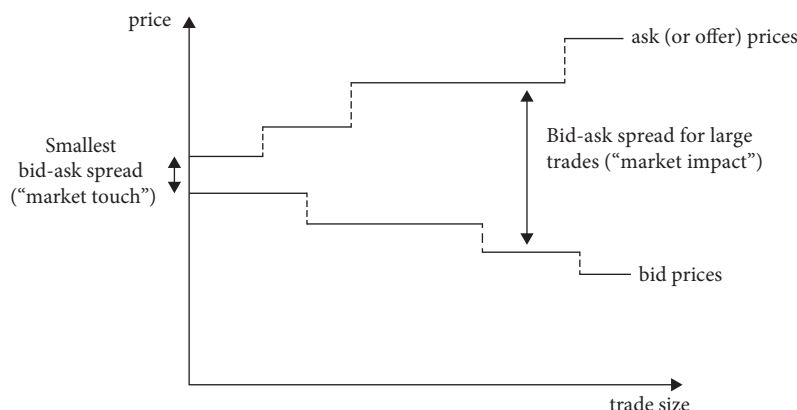


Figure 1.7 Dealer market quotes for various trade sizes.

good or important clients with better prices than other clients. For instance, in the U.S. market for municipal bonds (“munis”), dealers offer better prices to institutional than to retail investors, because institutions trade larger amounts, trade more frequently, and have greater bargaining power. Thus, the network of relationships of dealers and clients is an important determinant of the prices at which they trade in dealer markets (see Di Maggio et al. 2017 for evidence).

Moreover, bargaining may also speed up execution. For instance, an investor with a large order can ask the dealer to quote the price at which he is willing to take the entire order. Typically, this may be worse than the price the trader would obtain by splitting the order among several dealers over time, but it guarantees immediate execution of the full order. This is important to some traders, such as arbitrageurs who must take long and short positions simultaneously in different markets. Thus speed of execution constitutes an additional dimension of market liquidity, sometimes no less important than the cost of trading itself (i.e., the bid-ask spread).⁶

In the past, negotiations between dealers and their clients happened via voice (e.g., telephone) or messaging systems. In today’s OTC markets, they have been partially automated with the emergence of Single Dealer Platforms (SDPs), such as “Autobahn” (Deutsche Bank) or “Barx” (Barclays), which are internet portals enabling traders to access prices from one dealer bank for a variety of assets.

⁶ The importance of speed of execution for investors partly explains why the introduction of the continuous dealer market SEAQ International in London following the 1986 “Big Bang” reforms attracted so much trading in continental European shares. The desire to offer comparably rapid execution also induced continental European exchanges to replace their batch auction system with continuous electronic LOBs (Pagano and Röell 1990).

Moreover, in recent years, due to regulatory pressures following the global financial crisis, new trading systems, called Request for Quotes (RFQs) or click-to-trade, have been introduced in the D2C segments of fixed income and currency markets. For instance, for certain derivatives contracts (liquid standardized Interest Rate Swaps and Credit Default Swaps) in the U.S. the Dodd-Frank Act mandates trade execution on Swap Execution Facilities (SEFs), i.e., electronic limit order books, or RFQ systems, and also requires centralized clearing of trades; a similar regulation was introduced in 2018 in the E.U. as part of the MiFIR directive. RFQ systems enable investors to request quotes from multiple dealers simultaneously, rather than doing so sequentially, to discover the best price. Not surprisingly, such systems increase competition between dealers and have lowered trading costs for their clients. For instance, the mandated introduction of SEFs for dollar-denominated interest rate swaps triggered a reduction in total execution costs of about U.S. \$7–11 million per day (Benos et al. 2021), a sizable improvement in liquidity.

1.2.3 Hybrid Markets

Many actual securities markets are hybrids comprising both a limit order platform and a dealer segment, or having a design that mixes features of the two types of market.

For instance, the NYSE has a mix of three different trading mechanisms that operate simultaneously for each stock:

1. An open-outcry market where floor brokers trading on behalf of other investors or on their own account bargain bilaterally.⁷
2. A dealer market with designated market makers (DMM; formerly known as “specialists”) for each stock. DMMs have some obligations (e.g., they must post quotes at the best prices for a minimum fraction of the time during the day and their bid-ask spreads cannot exceed a certain threshold). Examples of firms acting as DMMs are Citadel Securities LLC or Virtu Americas LLC.
3. An electronic LOB (“NYSE Arca”) for each stock, which allows investors to bypass the DMMs and floor traders.

⁷ In an open-outcry market, brokers physically meet on a floor to trade. For instance, a broker announces that he is willing to buy 100 shares (a market order) and other brokers respond by shouting ask prices (limit orders) at which they are willing to execute the order. These markets mainly trade derivatives such as futures and options; with the development of electronic trading platforms they are tending to vanish.

Likewise, European exchanges (such as Euronext, Deutsche Börse, and OMX Nasdaq) also rely on DMMs.⁸ In this case, DMMs are hired by issuers and commit to maintain their bid-ask spread within a contractually specified limit. In compensation, they receive periodic payments from issuers. Such contracts (unauthorized in the U.S.) can be viewed as a way for issuers to “buy” liquidity and thereby reduce their cost of capital (see Section 11.5.2).

Furthermore, in both the E.U. and U.S. equity markets, investors can trade bilaterally with dealers off-exchange rather than on the respective exchanges’ limit order markets. This practice is called “internalization” as it entails dealers executing the orders for their own account. In E.U. equity markets, the banks, broker dealers and high-frequency market makers that operate under the status of “Systematic Internalizers” (SIs) are required to publicly display bid and ask quotes for the most liquid stocks. For other stocks, they have no such obligations. Their market share of total trading volume in E.U. equity markets has risen to 19% in 2020 (ESMA 2021a).

In U.S. equity markets, internalizers (such as Citadel securities, Virtu Securities, G1 Execution Services, sometimes referred to as “wholesalers”) accounted for about 27% of total trading volume in 2018 (SEC 2020). A large fraction of internalized trades stems from “payment for order flow” (PFOF) plans. According to these plans, brokers privately route orders with certain characteristics (e.g., orders from retail investors) to dealers, rather than public markets, in exchange for a small fee per share.⁹ The dealers that pay for order flow commit to execute market orders at prices that match the best quotes or improve upon them. The large fraction of retail order flow in U.S. equity and options markets that is internalized has raised various concerns about retail investor protection, such as whether internalized orders receive the best prices, and whether migration of orders away from exchanges could harm liquidity. These concerns are analyzed in Sections 7.2.2 and 7.2.3.

The evolution of trading in corporate bond markets is another interesting case in point. The share of electronic trading (via RFQs or all-to-all trading platforms operated by brokers such as TradeWeb or MarketAxess) has been steadily increasing,¹⁰ a trend that has accelerated during Covid-19.¹¹ Thus, even though the corporate bond market remains a dealer market, it now features

⁸ Exchanges use different terminologies to refer to DMMs, even though functions are similar. For instance, DMMs are called “designated sponsors” by Deutsche Börse and “Animateurs” by Euronext.

⁹ PFOF was introduced in the 1990s in the U.S. and has always been controversial. The advent of zero commission brokers (like Robinhood) gave a new vigor to this controversy because these brokers earn the bulk of their revenues from payments for order flow.

¹⁰ For instance, O’Hara and Zhou (2021) document that MarketAxess Inc. (the dominant electronic trading platform in U.S. corporate bonds) increased its share of investment-grade corporate bond trading from 10% in 2010 to 17% in 2016.

¹¹ See “Bond trading finally dragged into the digital age,” *Financial Times*, February 21, 2021.

both a standard dealer segment based on bilateral trading and a new electronic segment where customers can simultaneously access the quotes of many dealers.

1.2.4 Brokers vs. Dealers

Dealers differ from brokers, such as Charles Schwab or Robinhood for U.S. retail investors or ICAP for interdealer trades on OTC markets. Brokers only channel the buy and sell orders of their clients to the market, irrespective of whether it is a limit order market or a dealer market, without acting as counterparties for these orders. Dealers, instead, are the counterparties of final investors, and so take inventory risk. Some securities firms (such as Goldman Sachs or J.P. Morgan) offer both services, and are accordingly known as broker-dealers.

A feature that brokers and dealers have in common is that both help final investors to carry out their trades. For this reason, brokers and dealers are often collectively referred to as the “sell side” of the securities industry, whereas final investors (households, institutional investors, firms, and government) are called the “buy side” (since they buy trading services from the sell side).¹² This distinction is important since the two sides often have opposing views on how trading should be organized: as we shall see, a change in trading organization often affects the distribution of trading gains between investors (the buy side) and intermediaries (the sell side).

1.2.5 Market Transparency and Market Data

A market’s degree of transparency is determined by the amount of trading information available to participants, both about posted quotes—pre-trade transparency—and realized transactions (transaction prices and traded quantities)—post-trade transparency. This information (“market data”) matters to traders, because it enables them to sharpen their estimates of securities’ values (as order flows and prices contain information; see Chapter 3) and devise better trading strategies. For instance, as explained in Section 1.2.2, in a dealer market an investor can generally get better terms if he observes all dealers’ quotes, and can thus save on the cost of searching for the best price. Here we briefly discuss this important dimension of market structure, leaving a more in-depth analysis to Chapter 8.

¹² It is worth emphasizing that these terms do not denote buyers or sellers of securities but of trading services, so that the buy side includes both buyers and sellers of securities.

Transparency varies considerably from market to market. The level of transparency can be very controversial, as it affects how trading gains are distributed between the sell side and the buy side. Intuitively, the demand for brokerage services and dealers' market power are both greater in opaque markets, since it is harder for final investors to identify all trading opportunities. Thus, the sell side is usually less keen on improvements in transparency than the buy side. Moreover, when information on quotes and trades ("market data") is available, it is generally costly, as trading platforms charge significant fees for accessing real-time data. Thus, transparency depends on not only the availability but also the cost of market data. In recent years, this cost has increased, triggering heated debates about the pricing and availability of market data between the trading platforms that provide the data and their users.¹³

In general, electronic limit order markets tend to be very transparent. In these markets, trading platforms provide real-time data feed and charge fees that increase with the granularity of the content.¹⁴ At the least granular level, they only provide the best bid and ask quotes and realized transactions (prices and quantities); at the most granular level, they disclose the entire limit order book, with the exception of hidden limit orders (see Section 1.2.1). In some cases, they even disclose the identities of the brokers submitting limit orders or executed market orders.¹⁵

Transparency can also vary across dealer markets, depending on whether quotes are displayed centrally through a single screen and whether they are firm or merely indicative. In the latter case, customers still have to contact each dealer directly to verify the actual prices at which a trade will be executed. Since dealer markets are fragmented (deals being struck with individual dealers), it is not easy to ensure that the best quotes are centrally displayed in real time. For example, some dealers may quote prices only on their own proprietary systems. It is even harder to ensure that completed trades and their prices are published promptly, as this requires dealers to report their trades equally promptly to a central market authority. In practice, dealers often oppose prompt publication of trades or try to circumvent rules requiring such publication.

¹³ See, for instance, "Higher data fees prompt backlash against U.S. equity exchanges," *Financial Times*, November 17, 2017; "European investors complain about soaring costs of data," *Financial Times*, April 4, 2019; or "U.K. regulator to review charges for financial data," *Financial Times*, January 11, 2022.

¹⁴ See Oxera (2019) for estimates of these fees for European exchanges.

¹⁵ The transparency of LOB markets may also change over time. For instance, until 2002 only the best bid and ask prices in the LOB for the stocks listed on the NYSE were disseminated to market participants. Since then, all limit orders have been displayed through the exchange's "OpenBook" system. Another example is Euronext, which terminated disclosure of the identities of the brokers submitting limit orders in 2001.

Computerized trading facilitates data storage and dissemination, but it does not necessarily entail greater transparency in the trading process, for at least two reasons. First, the physical interactions of traders in open-outcry or “floor” markets permit the transmission of a host of informal cues that escape even state-of-the-art electronic systems. If traders are physically present in the same room, their every word and gesture are visible, yielding insights into their trading strategies and the urgency of their trading intentions.¹⁶ Moreover, the electrification of securities trading has facilitated the dispersal of trading across multiple venues (see Section 1.4.3 and Chapter 7), resulting in a fragmentation of market data across these venues. This fragmentation makes it hard to piece together an accurate view of overall market conditions.

To address this issue, the U.S. Securities and Exchange Commission (SEC) required the creation of a consolidated system for quotes and transactions reporting in equity markets in 1975. Pursuant to this mandate, U.S. exchanges report trade and quote data to “Securities Information Processors” (SIPs), which disseminate consolidated data (the best bid and offer prices for a stock across all exchanges and last trade reports) to the public with a very small delay (of a few milliseconds) relative to exchanges’ proprietary data feed. In fixed income markets, mandatory reporting of trade prices and sizes (after a delay of fifteen minutes) began in 2002 for corporate bonds via the Trade Reporting and Compliance Engine (TRACE) operated by FINRA (Financial Industry Regulatory Authority) and in 2005 for municipal bonds (via the EMMA system of the MSRB), with a 15-minute delay. In contrast, at the time of this writing, there is still no consolidated tape in U.S. Treasury markets or in E.U. equity and bond markets (but E.U. regulators are considering various options to introduce such a tape).

1.3 Does Market Structure Matter?

The two main functions of a security market are to provide trading services for investors who wish to alter their portfolios, and to determine prices that can guide the allocation of capital by investors and firms. A market is efficient in performing these functions if it enables investors to trade quickly and cheaply (i.e., if it is liquid) and if it incorporates new information quickly and accurately into prices (informational efficiency). Trading rules affect market efficiency on

¹⁶ Coval and Shumway (2001) show that changes in the sound level on the floor of the Chicago Board of Trade’s 30-year Treasury bond futures forecast subsequent changes in the cost of transacting and volatility. Their interpretation is that the sound level reflects how anxious market participants are to trade at current prices.

both accounts and this book is in part devoted to explaining why this is so. As a preview, we illustrate this point with a few examples below.

Some studies compare trading costs (bid-ask spreads) in limit order markets and dealer markets for matched samples of stocks (i.e., stocks with similar characteristics). Typically they find that trading costs are higher in dealer markets, especially for small orders. For instance, Huang and Stoll (1996) report that, by several measures, trading costs for a sample of Nasdaq stocks, when Nasdaq was still a dealer market, were twice as high as for a matched sample of NYSE stocks, when the NYSE was still primarily a floor market. One possible explanation is that concentrating trading in a single marketplace (in this case the floor of the NYSE) improves liquidity. Chapter 7 explores why and when the consolidation of trading can increase market liquidity.

Changes in trading rules that foster competition among liquidity providers also result in a more liquid market. In early 1997, the SEC introduced new rules (the “Order Handling Rules”) that exposed Nasdaq market makers to competition from the general public. The Limit Order Display Rule forced dealers to execute or display any customers’ limit orders that offered better prices than their own quotes. This regulatory change (together with the “quote rule” that required dealers trading in multiple venues to make their best quotes available to the public) increased competitive pressures on Nasdaq and led to an immediate and substantial reduction in Nasdaq trading costs (Barclay, Christie, Harris, Kandel, and Schultz 1999).

Changes in market transparency also affect liquidity. For instance, Boehmer, Saar, and Liu (2005) analyze how measures of market liquidity changed when the NYSE started releasing information on limit orders in 2002. They document that this increase in transparency was followed by a drop in the price impact of market orders—a measure of illiquidity alternative to the weighted average bid-ask spread. Similarly, Goldstein, Hotchkiss, and Sirri (2007) and Edwards, Harris, and Piwowar (2007) show that the implementation of TRACE (discussed in Section 1.2.5) significantly reduced bid-ask spreads in U.S. corporate bond markets by increasing post-trade transparency.

Another example of the importance of trading rules is provided by the emergence of electronic trading in fixed income markets: O’Hara and Zhou (2021) document that trading costs are smaller for electronic trades than for voice trades in U.S. corporate bonds, even after controlling for trade size, especially for small trades (for which electronic trading is more prevalent). Electronic trading also had a beneficial spillover effect on the traditional segment of the market, its growth being associated with a decline in trading costs for both electronic and voice trading between 2010 and 2017. This spillover is partly due to the fact that electronic trading reduces search costs for dealers’ clients and thereby increases competition between dealers.

Market structure also affects the quality of price discovery. Green, Li, and Schürhoff (2010) find that in the United States the prices of Treasury issues react to macroeconomic news much faster than the prices of munis. As the market for Treasuries is more liquid and active, the authors suggest that this is responsible for the different speed of price discovery. In turn, munis' lack of liquidity largely stems from the lack of transparency for their market, as documented by Green, Hollifield, and Schürhoff (2007). Many other studies have explored the impact of market transparency on liquidity and price discovery, as the detailed analysis in Chapter 8 shows.

By affecting liquidity and price discovery, market structure also affects the cost of capital. In Chapter 10, we will see that changes that increase liquidity raise stock prices, as investors require a lower rate of return to invest in more liquid stocks. Similarly, Chapter 12 shows how more efficient price discovery is conducive to better investment decisions by companies.

1.4 Evolution of Market Structure

We have seen that trading platforms are organized in a wide variety of ways. Why do actual trading mechanisms differ so much? This depends largely on who decides the trading rules, which is to say that it depends on the governance of trading platforms. In the past, such governance was partly the result of historical accident, which created different “initial conditions” in the various countries, as Section 1.4.1 explains. In recent decades, however, the governance and organization of trading platforms, especially in equity markets, have tended to converge on one basic model. Section 1.4.2 argues that this convergence reflects the heightened competition in the securities industry, itself the product of the combined pressure of three forces: liberalization of international capital markets, securities regulation overhaul, and advances in information and communication technologies. But technological innovation has transformed security trading over and above its tendency to reinforce competition between trading platforms. Section 1.4.3 describes the enormous impact of technological progress on the automation of order generation, routing, and execution, and the lively debate on the pros and cons of this evolution.

1.4.1 Who Makes the Rules?

The design of trading rules is the outcome of the interplay between government regulation and self-regulation by the trading platforms, such as stock exchanges or trading networks. For instance, in the U.S. much of securities regulation is

designed in broad outline by the SEC for equity and bond markets, and by the Commodity Futures Trading Commission (CFTC) for derivatives markets, while it is implemented and specified in greater detail by the self-regulating organizations that govern the markets, such as the NYSE, the Nasdaq and the Financial Industry Regulatory Authority (FINRA). In the E.U., the securities market regulator is the European Securities Markets Authority (ESMA), in coordination with E.U. members' national regulatory agencies (e.g., the AMF for France or Consob for Italy).

The scope of self-regulation and the way it has been used have varied widely over time and across countries. From their inception, stock exchanges have differed very significantly in governance and organization. Many emerged from informal trading. For example, what is widely regarded as the first modern exchange, the Amsterdam Stock Exchange, emerged at the start of the seventeenth century, when trading in the transferable shares of the Dutch East India Company started immediately upon issue in 1602. Trading was at first concentrated outdoors around the Nieuwe Brug in central Amsterdam (in foul weather, the nearby St. Olof's chapel was used), moving in 1611 to a purpose-built centralized merchants' exchange, created to facilitate public trading not just in shares but also in other financial instruments, goods, and insurance (see Petram 2011, for a detailed description of seventeenth-century Dutch share trading). Similarly, the London Stock Exchange (LSE) developed from informal trading that took place around coffee shops in the city, where in 1698 John Casting first published lists of stock prices, entitled "The course of the exchange and other things." Not until 1801 did the exchange turn into an official, regulated stock exchange. The precursor to the NYSE was created in 1792 by a group of stock brokers who signed the Buttonwood Agreement under a buttonwood tree on Wall Street.

Many continental European exchanges, however, were created at the initiative of government authorities and were generally regulated and managed by public agencies, such as local chambers of commerce. The Paris Stock Exchange was established by an order of the Royal Council of State in 1724. By the middle of the nineteenth century, trading was conducted by government-appointed *agents de change* who shouted out prices on the floor and could not trade for their own account: they were strictly brokers, not dealers. In several cases, the exchanges were founded for the express purpose of forming a market specifically for state-issued bonds, not shares or even corporate bonds. This was the case of the Vienna Stock Exchange, founded in 1771 under Empress Maria Theresa, and the Milan Stock Exchange, established in 1808 by a Napoleonic decree and used to trade only government bonds for its first half-century.

Thus, the constituencies that created stock exchanges were quite different, and the differences in governance were persistent. As late as the 1980s, most

exchanges were still governed much as they had been at the turn of the twentieth century. These differences in governance shaped the rules of the exchanges, in an illustration of the more general principle that the structure of securities markets is affected by the interests of their controlling constituencies. Depending on the relative power and importance of domestic and foreign intermediaries, institutional investors, and issuers in the share ownership structure, different exchanges pursue different policies. For instance, an exchange that is strictly controlled by a cartel of domestic intermediaries will be more inclined to impede remote access for foreign investors, and more generally will be reluctant to revise the trading system to the detriment of local intermediaries, as in the case of the LSE's protracted resistance to the introduction of an LOB and the NYSE members' opposition to ending floor trading.

More generally, the interests of intermediaries often conflict with those of issuers and investors: the former tend to favor trading rules that protect their rents at the expense of the latter. Chapters 7 and 8 show that such conflicts are a common feature in policy debates about market fragmentation and transparency.

By the same token, changes in the institutional form and governance of exchanges can impact the structure of their markets. Such changes have been frequent in recent years, as most exchanges have been transformed from essentially private clubs (such as the LSE) or semi-public entities (such as the Paris Bourse) into for-profit firms, in which the largest stakes are usually held by major financial intermediaries such as banks, hedge funds, and brokerage houses. The LSE, Deutsche Börse, Euronext, Nasdaq, Hong Kong Stock Exchange, and Toronto Stock Exchange all went public in 2000, while several other exchanges in continental Europe had done so in the late 1990s. The NYSE followed suit in 2006.

1.4.2 Competition between Exchanges

As just noted, from sharply divergent initial conditions, stock exchanges have converged on a similar status as publicly listed companies, through either the privatization of the government-controlled exchanges or the demutualization of those owned and controlled by groups of intermediaries. As for-profit firms, their choices regarding market structure are naturally more and more geared to maximizing shareholders' profits.¹⁷

¹⁷ This organizational form does not prevent deviations from the sole pursuit of profit maximization, where there are private benefits of control. In particular, if the stock exchange is controlled by sell-side firms, these may distort choices regarding market organization so as to serve their own

Table 1.1 Breakdown of Stock Exchange Revenues in 2014 (percent)

Source of Revenue	
Cash and capital markets trading	26
Derivatives and OTC markets trading	22
IT and market data services	19
Post-trade services	16
Financial income and other revenues	9
Listing and issuer services	8

Source: “Changing Business Models of Stock Exchanges and Stock Market Fragmentation,” OECD report, 2016.

To understand how this profit orientation altered their choices, note that trading revenues account only for a fraction of the total profits of an exchange (see Table 1.1). Other important sources of revenue are listing fees (paid by issuing firms who list their shares on the exchange), post-trade trading services (payments by traders to clear and settle their trades), and the sale of market data (to investors and other interested parties). For instance, in 2019 Nasdaq reported revenues from the sale of market data and technologies to use and access these data (e.g., co-location) totaling \$779 million (versus only \$414 million in 2014; see Ernst, Sokobin, and Spatt 2020). The sale of market data is important for European exchanges as well: in 2019 the sale of market data accounted for 11% of the LSE’s annual revenue and 19% of Euronext’s.¹⁸

The various services sold by exchanges are complements. Hence, decisions on market structure are made not only considering the impact on trading volume but also with an eye to their effect on listing and data sale revenues. For instance, an exchange may be willing to charge low listing fees in order to attract many issuers and so earn large trading revenues. Or one that operates its own clearing and settlement system may charge low trading fees in order to capture trading flows and charge large clearing and settlement fees.

In recent years, competition has been increasing in all of these business lines. The change in governance has been driven largely by this evolution: the exchanges became publicly listed companies so as to gain the flexibility needed to compete with alternative venues, both at home and abroad. The intensification of competition between trading platforms is the result of three forces:

interests. For instance, if their in-house systems are competing with the exchange’s platform in the provision of trading services, they may oppose changes in trading organization that increase the market share of the platform.

¹⁸ Exchange operators do not necessarily sell all these services. For instance, the Cboe does not provide listing services.

1. The removal of barriers to international capital flows has increased firms' propensity for initial public offerings (IPOs) on foreign markets or cross-listing of their shares, and prompted increased cross-border trading by investors. For instance, investors can currently trade many French blue-chip stocks not only on Euronext Paris but also on foreign markets: multilateral trading facilities (MTFs), such as Cboe, or exchanges where the stocks are cross-listed (such as Deutsche Börse and the LSE).
2. Changes in the securities market regulations have also played a role. In the United States, the increased fragmentation resulting from the proliferation of electronic communication networks (ECNs) provided major impetus for creating a new regulatory framework, Regulation National Market System (Reg NMS), in 2005. A major goal of Reg NMS is to organize and facilitate competition between trading platforms. For instance, its order protection rules oblige trading platforms to re-route incoming market orders to the platform that posts the best price at the moment the order is received. Clearly, this rule intensifies the competition for order flow among platforms, and among the liquidity suppliers that operate on each platform (see Chapter 7). In Europe, the Markets in Financial Instruments Directive (MiFID), which went into effect in 2007, also increased competition for order flow between trading platforms. In particular, it abolished the "concentration rule" by which member states could oblige investors to route their orders to the national market. As a consequence, it spurred the creation of many new platforms, operating Europe-wide limit-order books, such as Chi-X (launched in 2007), Turquoise, NASDAQ OMX Europe, Bats Europe (all launched in 2008), and NYSE Arca Europe (launched in 2009).¹⁹
3. Technological advances in information and communication technologies for securities trading have greatly facilitated entry into the market for the provision of trading services. The cost of setting up an electronic platform such as a LOB is now extremely low, which facilitated the creation of new electronic platforms challenging incumbent markets in the early '90s. For instance, one of the very first electronic limit order market, Island (subsequently renamed INET), emerged at the end of the '80s and challenged Nasdaq (which eventually acquired it). Other such platforms include Archipelago (now part of NYSE), Instinet and BATS (now part of Cboe). Moreover, with smart order-routing technologies (SORs) brokers can easily split orders across markets to get the best prices. Such technologies have lowered the cost of searching for the best price across trading

¹⁹ Chi-X and Bats no longer exist and have been acquired by the Cboe.

platforms and so made it easier for exchanges to attract trading business by offering narrow bid-ask spreads and low fees.

Intensified competition for order flow has forced trading platforms to reduce their trading fees. For instance, in 2003, the LSE announced that it would introduce a trading platform for Dutch stocks, EuroSETS. This was spurred by the Dutch brokerage community as a way to lower Euronext trading fees, and in fact Euronext slashed its fees by 50% in early 2004 (Foucault and Menkveld 2008). As a result, trading fees per share are now very small in equity markets (see Chapter 7 for more details).

Another benefit of competition is that it prompted the incumbents to overhaul trading systems and innovate. For instance, for long the NYSE relied heavily on a seriously outdated floor-based system, totally unable to match the execution speed of electronic trading. That trading system gave an informational advantage to members with a seat on the floor; they thus had a vested interest in retaining floor trading. But as competition from new electronic limit order book markets (e.g., BATS) eroded the NYSE's market share, this position became untenable. As a defensive move, in 2006, the NYSE acquired a rival electronic order market, Archipelago, renamed it NYSE Arca, and made it the core of its trading system. By 2009, Arca was processing four-fifths of the NYSE's total trading volume. As of 2017, there were still 205 floor brokers on the NYSE, but they mainly operated at the opening and closing sessions, according to the NYSE website.

Faced with such competition (whether actual or merely potential), the incumbent markets have also reacted by merging to achieve economies of scale and capitalize on liquidity externalities (see Chapter 7). For instance, in 2000 the Paris Bourse merged with the Amsterdam Stock Exchange and the Brussels Stock Exchange (subsequently joined by the Lisbon Stock Exchange), to form Euronext. Then, in 2007, Euronext in turn merged with the NYSE (giving rise to NYSE Euronext), enabling these exchanges to cut overhead costs and build up volume and liquidity. In 2014 the Intercontinental Exchange (ICE) acquired NYSE Euronext, and spun it off as an independent publicly listed company. Since then, Euronext has also acquired the Dublin, Oslo, and Italian Stock Exchanges, in 2017, 2019, and 2021, respectively.

At the time of writing, consolidation has given rise to five mega exchange operators, namely the Intercontinental Exchange, the Chicago Mercantile Exchange, the LSE, Deutsche Börse, and Nasdaq.²⁰ They have a wide scope of activities, such as the provision of trading and post-trading services in various

²⁰ These exchange operators account for about 50% of the total revenues (\$35 billions in 2019) in their industry. See "How exchange operators have grown bigger and bigger," *Financial Times*, December 12, 2020.

asset classes and the design and provision of financial indexes. Their business model is increasingly similar to that of big tech companies, in the sense that they use algorithms to match buyers and sellers of securities and monetize the data generated by this process. To boost revenues from market data (which are proportional to trading volume), they have incentives to reduce their trading fees, and in fact trading fees in electronic limit order markets have declined considerably over time. Moreover, they have acquired providers of data and analytics. For instance, the ICE acquired Interactive Data in 2015 for \$15 billion, and the LSE acquired Refinitiv (formerly Thomson-Reuters) in 2020 for \$27 billion.

1.4.3 Automation

As we have seen, technology has intensified competition between trading platforms. But the effects of information processing and communication technology on securities trading go well beyond this: over the last half-century technology has completely reshaped the trading process. In the past, securities were bought and sold on trading floors where brokers were in charge of matching the buy and sell orders they received from their clients. But securities exchanges progressively replaced or complemented their trading floors with computerized trading systems. By the turn of the century, 101 out of 120 countries already had electronic trading, and 85 of them no longer had floor trading (Jain 2005). Today, most exchanges use electronic LOBs, and orders are routed via high-speed fiber optic lines.

For the NYSE, this evolution began with the DOT system in 1976, which allowed electronic submission of market orders of up to one hundred shares. Upon reaching the trading floor, the electronic DOT orders for a stock were manually executed by the designated market maker (so called specialist) for this stock. But the NYSE did not become a fully electronic exchange until 2006 with the introduction of the NYSE Hybrid system. By contrast, such exchanges as the Toronto Stock Exchange and the Paris Bourse went over to fully electronic LOB markets quite early (1977 and 1986, respectively). Their trading systems were the blueprint for others.

This technological innovation has allowed traders (brokers, institutional investors' trading desks, proprietary trading firms, etc.) to automate the order submission process, giving rise to so-called *algorithmic trading* (see Chapter 9 for a detailed analysis). Algorithmic trading refers to the use of computers for submitting market and limit orders, and order cancellations according to codified rules ("algorithms"). These algorithms have often been developed by brokers to optimize the placement of large buy or sell orders by institutional investors (e.g., to minimize their trading costs, as discussed in Chapter 2).

Some firms specialize in a particular form of algorithmic trading, called *high-frequency trading* (HFT), which is characterized by the very high speed at which computers submit new orders in reaction to relevant events such as business news reports or changes in the state of limit order books. For instance, the SEC defines *high-frequency traders* (HFTs) as proprietary traders who use “extraordinary high speed and sophisticated computer programs for generating, routing, and executing orders.”

Trading speed is often measured by *trading latency*: the time lapse between the moment an electronic message is sent to a trading platform by a trader (for example, a market order to buy shares or a cancellation of a limit order) and the moment feedback is received from the trading platform (for example, on the status of the order). Latencies have become extremely short in electronic securities markets, close to physical limits. For instance, the average exchange-to-trader latency (the average time elapsed between the moment a message is sent by an exchange and the moment it is received by the firm) is 31 microseconds on average for messages about the SPDR S&P 500 Exchange Traded Fund (ETF), whose NYSE ticker code is SPY (Baldauf and Mollner 2020).

High-frequency trading firms (such as Citadel, Virtu Financial, Flow Traders, and Jump Trading) have made large investments to minimize trading latency. For instance, in 2013 alone, the Tabb Group estimated the investment in fast trading technologies at \$1.5 billion, twice the amount invested in 2012. Typically, these investments went into establishing fast communication lines between markets, either using fiber optics or microwave transmission, which is 30% faster than fiber optic lines. The first microwave network linking Chicago and New York became operational in 2010. It is used by arbitrageurs to obtain fast information on quotes for index futures traded on the Chicago Mercantile Exchange (CME) and for corresponding ETFs traded on platforms located in the New York area (e.g., NYSE, Cboe, or DirectEdge). Microwave transmission has reduced latency between Chicago and New York to 8.02 milliseconds, close to the theoretical lower bound (about 7.9 milliseconds). Investment by the proprietary trading desks of HFTs is often compared to an arms race, as the purpose is not speed per se but outrunning competitors (see Section 9.4).

Trading platforms have catered to the specific trading needs of high-frequency traders, by increasing the amount of market data made available to them and decreasing their own operating latencies. In fact, they are competing in the speed of disclosure of information on trades and quote updates so as to attract HFT volume and to sell quote information to them. One way they do so is by offering *co-location* services, that is, by renting rack space in their data centers. Co-location enables HFTs to run their algorithms in very close proximity

to exchanges' matching engines, so as to race ahead of their competitors in accessing information on trading opportunities and exploiting them.

As a result of these developments, by now HFT represents a significant fraction of the trading activity in electronic markets. For instance, according to ESMA (2021b), in 2018–19 HFT amounted to roughly 60% of total trading volume in European equity markets, and non-HFT algorithmic trading represented an additional 20%. Moreover, HFT has contributed to a very significant increase in the ratio of orders to trades in electronic markets, forcing electronic platforms to increase their processing capacity in the face of a surfeit of limit order cancellations and resubmissions. The dominance of algorithmic and high-frequency trading in today's securities markets is attracting intense public scrutiny: its impact on liquidity, price discovery, volatility, and risk is widely debated, as we shall see in Chapter 9.

1.5 Further Reading

Harris (2003) provides a detailed description of the trading mechanisms and the organization of the securities industry in the United States, and Lyons (2001) describes the structure and functioning of currency markets and offers a theoretical and empirical analysis of their operation. Pagano and Röell (1990, 1993) describe how European trading systems started changing in the 1980s under the pressure of deregulation, competition, and technological innovation. Lee (2000) analyzes in depth the governance of exchanges and their sources of revenue.

Oxera (2019) and SEC (2020) provide accounts of the more recent evolution of the market structure of European and U.S. equity markets, respectively. Description of the evolution of trading mechanisms in currency markets are provided by Chaboud, Rime, and Sushko (2022), Schrimpf and Sushko (2021), BIS (2016), BIS (2020). For fixed income markets, see Bech, Illes, Lewrick, and Schrimpf (2016), and Bessembinder, Spatt, and Venkataraman (2020). Mizrach and Neely (2007) and U.S. Treasury Department (2017) describe the structure of the U.S. Treasury bond market. Duffie et al. (2022) (Chapter 4) provide an overview of trading automation on various markets and its effects.

Pagano and von Thadden (2004) and Dunne, Moore, and Portes (2006) describe and analyze the European government bond market. Biais, Declerck, Dow, Portes, and von Thadden (2006) analyze the European corporate bond market, with special attention to its transparency. European Commission (2017) provides an overview of the evolution of European corporate bond markets.

1.6 Exercises

- 1. Call auction.** Graph total market demand and supply curves in {price, quantity} space for a call auction market where the following orders are submitted to a central auctioneer:

Limit orders to buy: 100 shares at \$3.00, 200 shares at \$4.00, 200 shares at \$3.50, and 500 shares at \$2.50.

Limit orders to sell: 500 shares at \$5.00, 600 shares at \$3.00, and 500 shares at \$4.00.

Market orders to buy: a total of 500 shares.

Market orders to sell: a total of 200 shares.

What is the market clearing price? What quantity of stock is traded? Are all orders that are executable at the market clearing price fully filled?

- 2. Continuous order-driven market.** Now suppose that the above orders arrive on the market over time, in the order of arrival that is listed above (that is, at time $t = 1$ the limit order to buy 100 at \$3.00 is submitted, at time $t = 2$ the limit order for 200 at \$4.00, and so on, continuing until time $t = 9$, when the market order to sell 200 arrives). Track the state of the LOB (show it after each new order has arrived and any transactions are triggered, for $t = 0, \dots, 9$, in the trading screen format of Figure 1.2) and the time, price and quantity of any transactions that take place. Record the dollar bid-ask spread, that is, the difference between the lowest ask and the highest bid, in the continuous market as it evolves from $t = 5$ onwards.
- 3. Comparison: efficiency and market presence.** Consider again the two markets described in questions 1 and 2. Assume that the limit order prices are equal to the order placer's valuation for the block of shares submitted in the order, and think of market orders as placed by agents whose valuation is well outside (above for buyers, below for sellers) the relevant range of trading prices. Which market is Pareto efficient, in the sense that at the end of the trading day there is no pair of agents who could both benefit by trading with each other (i.e., after $t = 9$ in the continuous order-driven market)? Intuitively, why?
- 4. Effect of limit order execution on the LOB.** Consider a limit order book (LOB) market. Suppose that the best limit orders on the LOB are:

Bid		Ask	
Shares	Price	Shares	Price
100	90.10	300	91.40
200	89.80	200	92.00
300	89.20	500	92.10
500	89.00	300	93.40

- a. Suppose a sell order of 800 with a limit price of 89.10 reaches the market and is executed. What are the new ask and bid prices after the execution of the order?
 - b. How do the bid-ask spread defined by the BBO (best bid and offer) and the market depth available at the BBO change relative to the initial LOB? Explain briefly why.
5. **Effect of limit and market orders execution on the LOB.** Suppose that the best limit orders in the limit order book are:

Bid		Ask	
Shares	Price	Shares	Price
100	59.60	300	61.40
200	59.50	200	61.70
1,000	59.20	500	62.00
500	59.00	300	62.40

Suppose a sell order of 800 with a limit price of 59.30 reaches the market and is executed. Immediately afterwards, a market buy order of 600 reaches the market and is executed.

- a. Explain how these two orders modify the state of the LOB and show what is the final state of the LOB.
 - b. Is the final quoted bid-ask spread larger or smaller than the initial one? What is the intuitive reason?
 - c. Suppose the sequence of the two orders were the opposite: first the market buy order of 600 and then the limit sell order of 800. Would the final state of the LOB be the same, and would the final bid-ask spread be larger or smaller than the initial one in this case? What is the intuitive reason?
 - d. Does this illustrate any general principles? Which ones?
6. **Call auctions and price impacts.** Consider a stock that trades in a call auction market. Just before the call auction clears, the following orders are submitted to the market:
1. **Sell limit orders:** 5 shares at prices \$50, \$52, \$53, \$54, \$55, \$56, \$57, and \$58.
 2. **Buy limit orders:** 20 shares at \$55, 20 shares at \$54, 5 shares at \$53, 20 shares at \$52, 20 shares at \$51, and 20 shares at \$50.
 - a. If no additional order is submitted, what is the clearing price for the stock? How many shares are traded at this price? What is the bid-ask spread after the call auction?
 - b. Suppose that you receive negative information about the stock, which leads you to expect the value of the stock to be \$50 in 1 hour.

To profit from this information, you consider posting a sell limit order at \$50. You ponder how many shares, x , you should offer, considering increments of 10 shares (10 shares, 20 shares, etc.). Your objective is to maximize your total expected profit, which is equal to the number of shares you sell in the call auction times the difference between the clearing price (accounting for your order) and your estimate (\$50) of the value of the asset in 1 hour. You know that other traders have placed limit orders as specified above. If your limit order is tied with other limit orders at the clearing price, the orders placed by other traders will be executed first, because time priority is enforced. What is the optimal size x of your limit order, and what is your expected profit from this order? Justify your answer.

- c. You learn that another seller is about to place a sell limit order at \$50. Without redoing any calculations, should you consider (i) increasing the size of your sell limit order, (ii) leaving it unchanged, or (iii) reducing it? Explain whether and how your answer depends on the time priority and the size of this limit sell relative to yours.
7. **Continuous auctions.** The table below shows the limit order book of Nestlé at 9:30 a.m. on October 23, 2018, on Cboe (a trading platform for European stocks):

Bid		Ask	
Shares	Price	Shares	Price
1,050	83	1229	83.1
450	82.98	1,183	83.08
470	82.96	1,255	83.6
718	82.94	1,474	83.04
669	82.92	383	83.02

- a. What is the quoted bid-ask spread for Nestlé at 9:30 a.m.?
- b. If you place a buy market order for 3,000 shares of Nestlé at 9:30 a.m., how much will you pay overall to acquire these shares? What is the effective bid-ask spread on this trade?
- c. If you place a sell market order for 2,000 shares, what will be the quoted bid-ask spread and the number of shares offered at the best quotes after this transaction?
- d. If you place a buy limit order for 2,000 shares at a price equal to €82.99, what will be the new state of the limit order book after the arrival of this order?

- e. The trader who has placed a sell limit order for 383 shares at €83.02 is a dealer. Suppose that Nestlé unexpectedly releases earnings forecasts that are higher than expected at 9:31 a.m. and that, given this news, the new estimate of Nestlé stock price (per share) is €84. If the dealer does not update his quotes, explain what you would expect other market participants to do, and what would be the resulting impact on the dealer.